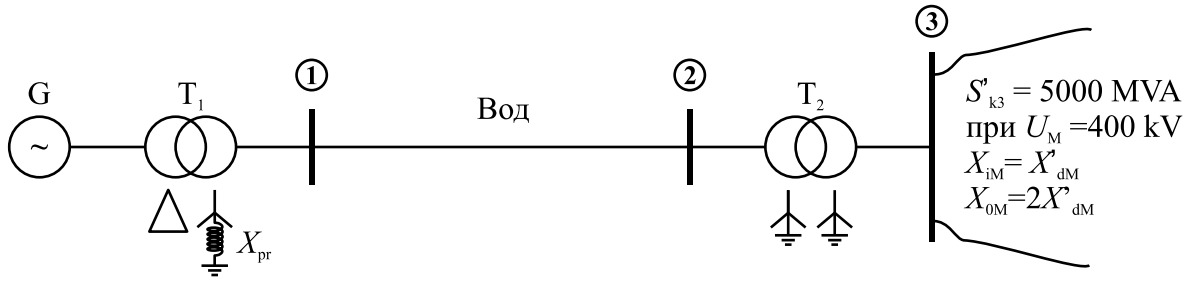
Задатак за вјежбу,
Бања Лука, 13. април 2020. године

Задатак 3

На слици 1 је дат једноставан електроенергетски систем. Одредити потребну индуктивност пригушнице у звјездишту трансформатора T_1 да би однос струја једнополног и трополног кратког споја на сабирницама 1 износио 0,9.



$S_{nG} = S_{nT1} = 200$ MVA
$U_{nG} = 10{,}5$ kV
$m_{T1} = 10{,}5/231$ kV/kV
$x'_{dG} = x_{iG} = 20$ %
$x_{T1} = 12$ %

$U_{nV} = 220$ kV
$x_V = 0{,}4$ Ω/km
$x_{0V} = 1{,}2$ Ω/km
$L_V = 100$ km

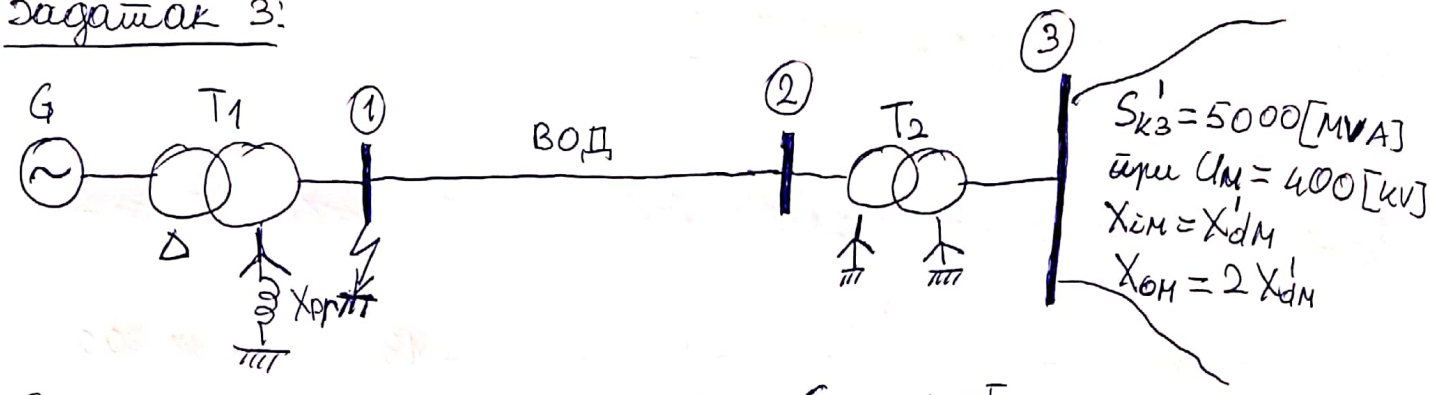
$S_{nT2} = 200$ MVA
$m_{T2} = 220/400$ kV/kV
$x_{T2} = 10$ %

Слика 1 – Посматрани дио електроенергетског система

ЗАДАТАК ЗА ВЈЕЖБУ

13.04.2020.

Задатак 3:



$S_{nG} = S_{nT1} = 200 [MVA]$ $\quad U_{nV} = 220 [kV]$ $\quad S_{nT2} = 200 [MVA]$

$U_{nG} = 10.5 [kV]$ $\quad x_V = 0.4 \left[\frac{\Omega}{km}\right]$ $\quad m_{T2} = \frac{220}{400} \left[\frac{kV}{kV}\right]$

$m_{T1} = \frac{10.5}{231} \left[\frac{kV}{kV}\right]$ $\quad x_{0V} = 1.2 \left[\frac{\Omega}{km}\right]$ $\quad x_{T2} = 10\%$

$x'_{dG} = x_{iG} = 20 [\%]$ $\quad L_V = 100 [km]$

$x_{T1} = 12 [\%]$

- Вриједности параметара анализираног ЕЕС-а сведени на напонски ниво 220 [kV] (на тој страни се десио квар):

• генератор:

$$X'_{dG} = \frac{x'_{dG\%}}{100} \cdot \frac{U_{nG}^2}{S_{nG}} \cdot \frac{1}{m_{T1}^2} = \frac{20}{100} \cdot \frac{10.5^2}{200} \cdot \left(\frac{231}{10.5}\right)^2 = 53.3610 [\Omega]$$

$$X_{iG} = X'_{dG} = 53.3610 [\Omega]$$

• Тф-и:

$$X_{T1} = \frac{x_{T1\%}}{100} \cdot \frac{U_{nT1}^2}{S_{nT1}} = \frac{12}{100} \cdot \frac{231^2}{200} = 32.0166 [\Omega]$$

$$X_{T2} = \frac{x_{T2\%}}{100} \cdot \frac{U_{nT2}^2}{S_{nT2}} = \frac{10}{100} \cdot \frac{220^2}{200} = 24.2 [\Omega]$$

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• Вод:

$$X_V^d = X_V^i = x_V L_V = 0.4 \cdot 100 = 40\,[\Omega]$$

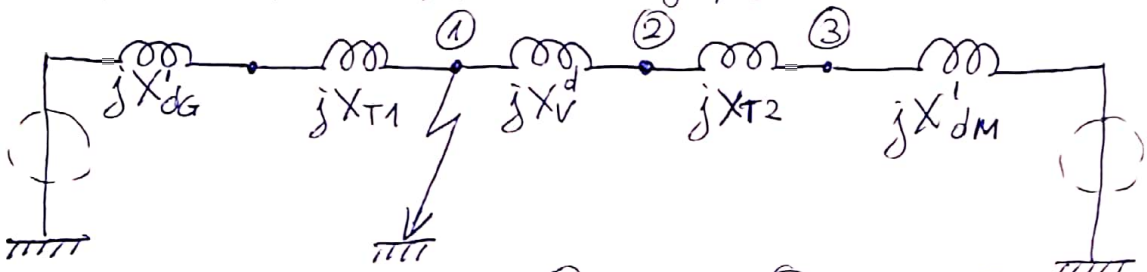
$$X_V^0 = x_{0V} L_V = 1.2 \cdot 100 = 120\,[\Omega]$$

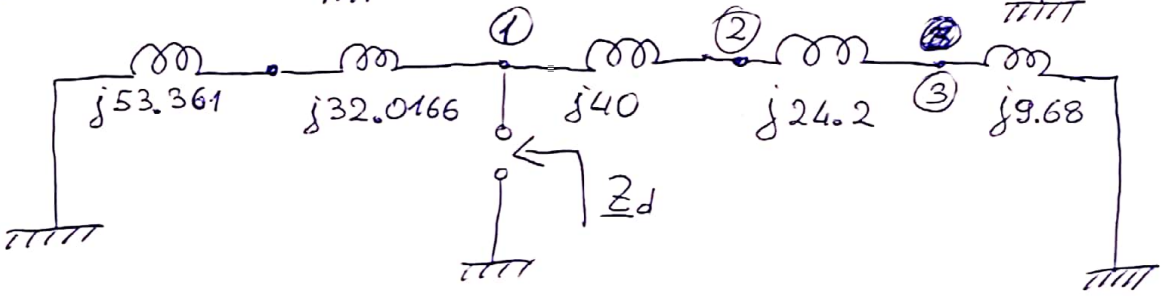
• јака мрежа:

$$X'_{dM} = \frac{U_N^2}{S'_{K3}} \cdot m_{T2}^2 = \frac{400^2}{5000} \cdot \left(\frac{220}{400}\right)^2 = 9.68\,[\Omega]$$

$$X_{0M} = 2X'_{dM} = 2 \cdot 9.68 = 19.36\,[\Omega]$$

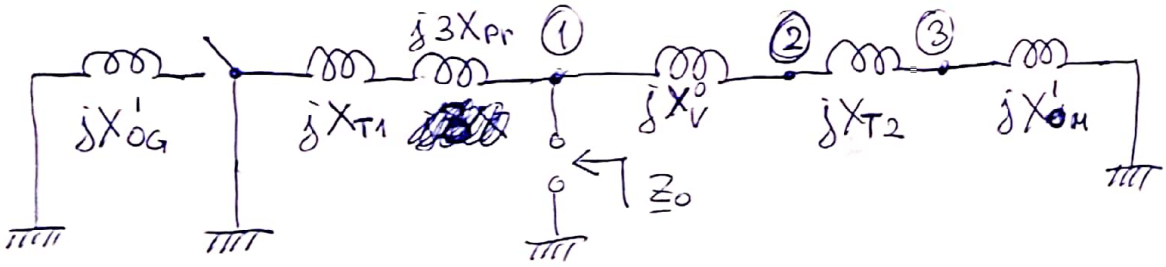
- Еквивалентна шема за директни/инверзни редослед

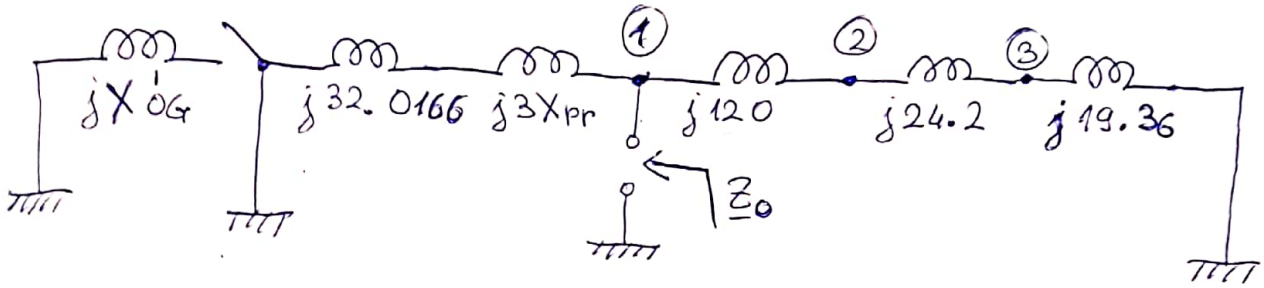




$$\underline{Z}_d = \underline{Z}_i = \frac{j(53.361 + 32.0166) \cdot j(40 + 24.2 + 9.68)}{j(53.361 + 32.0166 + 40 + 24.2 + 9.68)} = j39.6069\,[\Omega]$$

- Еквивалентна шема за нулти редослед





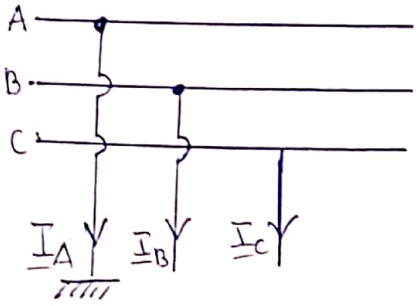
$$\underline{Z}_L = j(32.0166 + 3X_{pr})$$

$$\underline{Z}_D = j(120 + 24.2 + 19.36) = j163.56\ [\Omega]$$

$$\underline{Z}_0 = \underline{Z}_L \parallel \underline{Z}_D = \frac{j(32.0166 + 3X_{pr}) \cdot j163.56}{j(32.0166 + 3X_{pr} + 163.56)}$$

$$\cancel{Z_0 = (5236.635)} \qquad \underline{Z}_0 = \frac{j(5236.6351 + 490.68X_{pr})}{195.5766 + 3X_{pr}}$$

1PKS



$\Rightarrow$ j-не физичне отпадности

$\underline{U}_A = 0,\ \underline{I}_B = 0,\ \underline{I}_C = 0$

$$\begin{bmatrix} \underline{I}_d \\ \underline{I}_i \\ \underline{I}_0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} \underline{I}_A \\ \underline{I}_B \\ \underline{I}_C \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} \underline{I}_A \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} \underline{I}_A \\ \underline{I}_A \\ \underline{I}_A \end{bmatrix}$$

$$\Rightarrow \underline{I}_d = \underline{I}_i = \underline{I}_0$$

$$\begin{bmatrix} \underline{U}_d \\ \underline{U}_i \\ \underline{U}_0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} \underline{U}_A \\ \underline{U}_B \\ \underline{U}_C \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ \underline{U}_B \\ \underline{U}_C \end{bmatrix}$$

$$\underline{U}_A = 0 = \underline{U}_d + \underline{U}_i + \underline{U}_0$$

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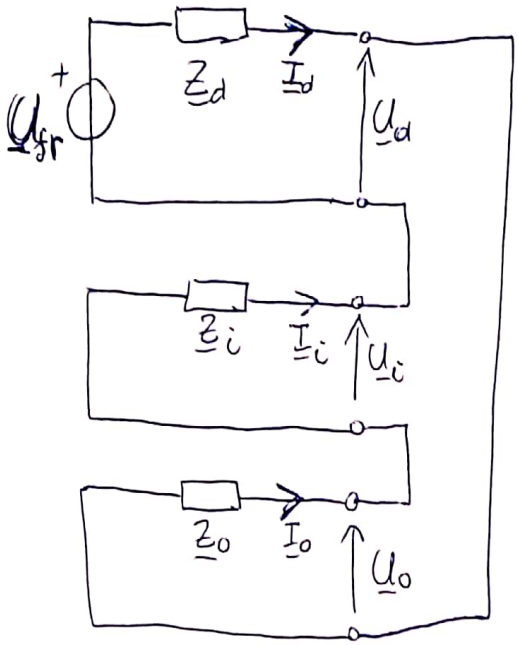


$$\underline{I}_d = \underline{I}_i = \underline{I}_0$$

$$\underline{I}_d = \frac{\underline{U}_{fr}}{\underline{Z}_d + \underline{Z}_i + \underline{Z}_0}$$

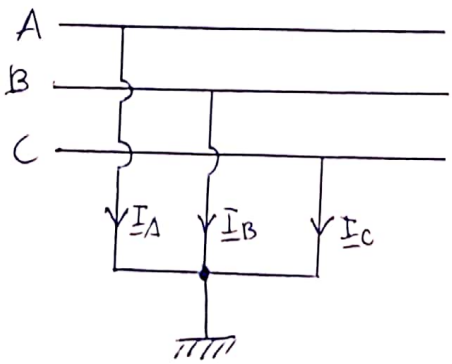
$$\underline{U}_d = \underline{U}_{fr} - \underline{Z}_d \underline{I}_d$$

$$\underline{U}_i = -\underline{Z}_i \underline{I}_i$$

$$\underline{U}_0 = -\underline{Z}_0 \underline{I}_0$$

$$\underline{I}_{1PKS} = 3\underline{I}_0 = \frac{3\underline{U}_{fr}}{\underline{Z}_d + \underline{Z}_i + \underline{Z}_0}$$

3PKS



$\Rightarrow$ ј-не физичке очитодности

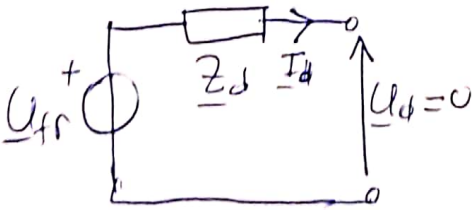
$\underline{U}_A = 0, \ \underline{U}_B = 0, \ \underline{U}_C = 0$

инверзни и нулти систем не егзистирају због симетрије квара

$\underline{I}_A = \underline{I}_A, \ \underline{I}_B = \underline{a}^2 \underline{I}_A, \ \underline{I}_C = \underline{a}\, \underline{I}_A$

$$\begin{bmatrix} \underline{I}_d \\ \underline{I}_i \\ \underline{I}_0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \underline{I}_A \\ \underline{I}_B \\ \underline{I}_C \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \underline{I}_A \\ \underline{a}^2 \underline{I}_A \\ \underline{a}\, \underline{I}_A \end{bmatrix} = \frac{1}{3} \begin{bmatrix} \underline{I}_A + \underline{I}_A + \underline{I}_A \\ 0 \\ 0 \end{bmatrix}$$

$$\underline{I}_d = \underline{I}_A$$



$$\underline{I}_{3PKS} = \frac{\underline{U}_{fr}}{\underline{Z}_d}$$

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$$\frac{|\underline{I}_{1pks}|}{|\underline{I}_{3pks}|} = 0.9$$

$$|\underline{I}_{1pks}| = \left|\frac{3\underline{U}_{fr}}{\underline{Z}_d + \underline{Z}_i + \underline{Z}_0}\right| \qquad \underline{I}_{3pks} = \left|\frac{\underline{U}_{fr}}{\underline{Z}_d}\right|$$

$$\left|\frac{\underline{I}_{1pks}}{\underline{I}_{3pks}}\right| = \left|\frac{\dfrac{3\cancel{\underline{U}_{fr}}}{\underline{Z}_d + \underline{Z}_i + \underline{Z}_0}}{\dfrac{\cancel{\underline{U}_{fr}}}{\underline{Z}_d}}\right| = \left|\frac{3\,\underline{Z}_d}{\underline{Z}_d + \underline{Z}_i + \underline{Z}_0}\right| = 0.9$$

$$\cancel{\left|\frac{3\cdot j39.6069}{2\cdot j39.6069 + \frac{195.5766 + 490.68X_{pr}}{195.}}\right|}$$

$$\left|\frac{3\cdot j39.6069}{2\cdot j39.6069 + \frac{j(5236.6351 + 490.68X_{pr})}{195.5766 + 3X_{pr}}}\right| = 0.9$$

$$\left|\frac{j118.8207}{j79.2138 + \frac{j(5236.6351 + 490.68X_{pr})}{195.5766 + 3X_{pr}}}\right| = 0.9$$

$$\left|\frac{j(23238.5485 + 356.4621X_{pr})}{j(15492.3656 + 237.6414X_{pr} + 5236.6351 + 490.68X_{pr})}\right| = 0.9$$

$$\frac{23238.5485 + 356.4621X_{pr}}{20729 + 728.3214X_{pr}} = 0.9$$

$$23238.5485 + 356.4621X_{pr} = 18656.1 + 655.9393X_{pr}$$

$$4582.4485 = 299.4772X_{pr} \Rightarrow \boxed{X_{pr} = 15.3015\,[\Omega]}$$

$$X_{pr} = 2\pi f L_{pr} \Rightarrow L_{pr} = \frac{X_{pr}}{2\pi f} = \frac{15.3015}{2\pi\cdot 50} \Rightarrow \boxed{L_{pr} = 48.71\,[mH]}$$

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